DERIVED COMPLETION COMMUTES WITH THE INTERNAL HOM IN EITHER ARGUMENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- For a finitely generated ideal the derived completion functor is given by the internal Hom from a fixed complex, hence has a right adjoint description. invoked at proof lines 26268-26270. Turns the completion of the internal Hom into the internal Hom of a tensor product with that fixed complex.
- The internal Hom satisfies the tensor-hom adjunction, so it can be moved from one argument to the other at the cost of tensoring. invoked at proof lines 26271-26274. Produces the chain of equalities converting the completion of the Hom of K and L into the Hom of K with the completion of L.
- The comparison map from a complex to its derived completion is an isomorphism after applying the internal Hom from the completion, by the defining adjunction of the completion. invoked at proof lines 26278-26285. Gives the second equality, allowing both arguments to be replaced by their completions.

Source reference. Original lemma 091V: sources/stacks-more-algebra/pinned-original-source.tex, source lines 26154–26154. Invoked at proof lines 26268-26270 to replace the completion by an internal Hom; every adjunction step performed afterwards is inside the counted span.

Source reference. Original lemma 0A65: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18682–18682. Used at proof lines 26271-26274 for the two displayed equalities converting the Hom into a Hom of the completed argument; the remaining adjunction argument is inside the counted span.

Lemma 0.2. *Let A be a ring and let $I \subset A$ be a finitely generated ideal. Let $K, L \in D(A)$. Then*

$$R\operatorname{Hom}_A(K, L)^\wedge = R\operatorname{Hom}_A(K, L^\wedge) = R\operatorname{Hom}_A(K^\wedge, L^\wedge)$$

Proof. By Lemma 091V we know that derived completion is given by $R\operatorname{Hom}_A(C, -)$ for some $C \in D(A)$. Then

$$\begin{aligned} R\operatorname{Hom}_A(C, R\operatorname{Hom}_A(K, L)) &= R\operatorname{Hom}_A(C \otimes^{\mathbf{L}}_A K, L) \\ &= R\operatorname{Hom}_A(K, R\operatorname{Hom}_A(C, L)) \end{aligned}$$

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

by Lemma 0A65. This proves the first equation. The map $K \to K^\wedge$ induces a map

$$R\mathrm{Hom}_A(K^\wedge, L^\wedge) \to R\mathrm{Hom}_A(K, L^\wedge)$$

which is an isomorphism in $D(A)$ by definition of the derived completion as the left adjoint to the inclusion functor. $\square$